

Department of Mathematics
Motilal Nehru National Institute of Technology Allahabad
Mathematics II (MAN12101)
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Assignment 1 (Unit I)

1. Let $V = \{(z_1, z_2, z_3) : z_1, z_2, z_3 \in \mathbb{C}\} = \mathbb{C}^3$ be the vector space over $\mathbb{C}$. Check whether the following subspace of V .
a. $W_1 = \{(z_1, z_2, z_3) : z_1 + z_2 = 0\}$ b. $W_2 = \{(z_1, z_2, z_3) : z_1 + z_2 = 1\}$ c. $W_3 = \{(z_1, z_2, z_3) : z_1 \in \mathbb{R}\}$
2. Show that the set of all real valued continuous functions $y = f(x)$ satisfying differential equations $y''' + 6y'' + 11y' + 6y = 0$ is a vector space over $\mathbb{R}$. Find a basis of this.
3. Let V be the set of all positive real valued continuous function f on $[a, b]$ such that $\int_a^b f(x)dx = 2$ with usual addition and scalar multiplication. Show that V is not a vector space.
4. Show that the set $V = \{(x, y, z) : x, y, z \in \mathbb{R} \text{ and } x + y = 11\}$ is not a subspace of $\mathbb{R}^3$.
5. Show that the set $V = \{(x, y, z) : x, y, z \in \mathbb{R} \text{ and } x + 2y + z = 0\}$ is a subspace of $\mathbb{R}^3$.
6. Consider the polynomials $p(x) = 1 + 3x + 2x^2$, $q(x) = 3 + x + 2x^2$, $r(x) = 2x + x^2$ in $P_2(x)$. Show that the set $\{p(x), q(x), r(x)\}$ is linearly dependent.
7. Determine conditions on the scalars a, b so that the set $S = \{v_1, v_2, v_3\}$ of vectors is linearly dependent, where $v_1 = [1 \ 3 \ 1]^T$, $v_2 = [1 \ a \ 4]^T$, $v_3 = [0 \ 2 \ b]^T$.
8. Show that the set $\{1, x, 1 + x + x^2\}$ is a linearly independent set of vectors in the vector space of all polynomials over $\mathbb{R}$.
9. Let V be the vector space of all polynomials of degree ≤ 3 . Determine whether or not the set $\{t^3, t^2 + t, t^3 + t + 1\}$ spans V ?
10. Show that the vectors $\{[1 \ 2 \ 3]^T, [3 \ 5 \ 7]^T, [5 \ 9 \ 13]^T\}$ and $\{[1 \ 2 \ 3]^T, [3 \ 5 \ 7]^T, [0 \ 1 \ 2]^T\}$ do not span $\mathbb{R}^3$.
11. Write $t^2 + t + 1$ as a linear combination of the elements of the set $S : \{3t, t^2 - 1, t^2 + 2t + 2\}$. Show that S is the spanning set for all polynomials of degree 2 and can be taken as its basis.
12. Let $V = \{(x_1, x_2, x_3, x_4) : x_1 - x_2 + x_3 - x_4 = 0\}$ be the subspace of the 4-dimensional vector space $\mathbb{R}^4$. Find a basis and dimension of the subspace V .
13. Let T be a linear transformation from $\mathbb{R}^4$ into $\mathbb{R}^3$, where $T \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} x + y + w \\ z \\ y + 2w \end{bmatrix}$. Find $\ker(T)$, $\text{ran}(T)$ and their dimensions.
14. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x + y \\ x - z \end{bmatrix}$. Find the matrix representation of T with respect to the ordered basis $x = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ in $\mathbb{R}^3$ and $y = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}$ in $\mathbb{R}^2$.

15. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$ be the matrix representation of the linear transformation T with respect to the ordered basis vectors $\mathbf{v}_1 = [1, 2]^T$, $\mathbf{v}_2 = [3, 4]^T$ in $\mathbb{R}^2$ and $\mathbf{w}_1 = [-1, 1, 1]^T$, $\mathbf{w}_2 = [1, -1, 1]^T$, $\mathbf{w}_3 = [1, 1, -1]^T$ in $\mathbb{R}^3$. Then, determine the linear transformation T .
16. Define two functions $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + y \\ 0 \end{bmatrix}$ and $S \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + y \\ xy \end{bmatrix}$. Determine whether T , S and the composite $S \circ T$ are linear transformations.
17. Let V be the vector space of 2×2 matrices, and W be the vector space of 3×2 matrices. Define the linear transformation $T : V \rightarrow W$ by $T \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = \begin{bmatrix} a + b & 2d \\ 2b - d & -3c \\ 2b - c & -3a \end{bmatrix}$. Find a basis for the range of T .
18. Let T be the linear transformation from the 3-dimensional vector space $\mathbb{R}^3$ into $\mathbb{R}^3$ itself satisfying the following relations: $T \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $T \left(\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$, $T \left(\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Then for any vector $x \in \mathbb{R}^3$, find the formula for $T(x)$.
19. Check the consistency of the following system of equation and if consistent then find the solution:
 (i) $2x - y + 3z = 9$, $x + y + z = 6$, $x - y + z = 2$, $x + y - z = 0$
 (ii) $x + y + z = 6$, $2x + 3y - 2z = 6$, $5x + y + 2z = 13$.
20. Find the values of λ and μ for which the system of equations $x + 2y + z = 6$, $x + 4y + 3z = 10$, $x + 4y + \lambda z = \mu$ has i. a unique solution, ii. infinite number of solution, iii. no solution.
21. Solve the following system of equations using Gauss elimination method.
- (i) $4x - 3y - 9z + 6w = 0$, $2x + 3y + 3z + 6w = 6$, $4x - 21y - 39z - 6w = -24$
- (ii) $2x + 4y + 6z = 22$, $3x + 8y + 5z = 27$, $-x + y + 2z = 2$